Bash Scripting Assignment

maintenance_assistant.sh | Full script, part 1 of 2

```
#!/bin/bash

echo "======================================"
echo "      System Maintenance Assistant"
echo "======================================"

# Part 1: Welcome Message
read -r -p "Enter your name: " NAME

if [ -z "$NAME" ]; then
    echo "Error: Name cannot be empty."
    exit 1
fi

echo "Hello, $NAME!"

# Part 2: Root Check
if [ "$EUID" -ne 0 ]; then
    echo "Error: This script must be run as root."
    exit 1
fi

echo "Root check passed."

# Part 3: Backup Folder Creation
CURRENT_DATE=$(date +%y-%m-%d)
BACKUP_DIR="/tmp/backup-$CURRENT_DATE"

if [ -d "$BACKUP_DIR" ]; then
    echo "Backup directory already exists: $BACKUP_DIR"
else
    mkdir -p "$BACKUP_DIR"
    echo "Backup directory created: $BACKUP_DIR"
fi

# Part 4: Source File Validation
read -r -p "Enter the source log file path: " SOURCE_FILE

if [ ! -f "$SOURCE_FILE" ]; then
    echo "Error: Source file does not exist."
    exit 1
fi

if [ ! -s "$SOURCE_FILE" ]; then
    echo "Error: Source file is empty."
    exit 1
fi

echo "Source file is valid."
```

Script and short explanation

maintenance_assistant.sh | Full script, part 2 of 2

```
# Part 5: Backup Process
BACKUP_FILE="$BACKUP_DIR/$(basename "$SOURCE_FILE").bak"

cp "$SOURCE_FILE" "$BACKUP_FILE"

if [ $? -eq 0 ]; then
    echo "Backup successful: $BACKUP_FILE"
else
    echo "Error: Backup failed."
    exit 1
fi

# Part 6: Service Control
SERVICE="httpd"

case "$1" in

    start)
        systemctl start "$SERVICE"
        echo "$SERVICE started."
        ;;

    stop)
        systemctl stop "$SERVICE"
        echo "$SERVICE stopped."
        ;;

    restart)
        systemctl restart "$SERVICE"
        echo "$SERVICE restarted."
        ;;

    status)
        systemctl status "$SERVICE" --no-pager
        ;;

    *)
        echo "Usage: $0 {start|stop|restart|status}"
        exit 1
        ;;

esac

echo "System maintenance task completed."
```

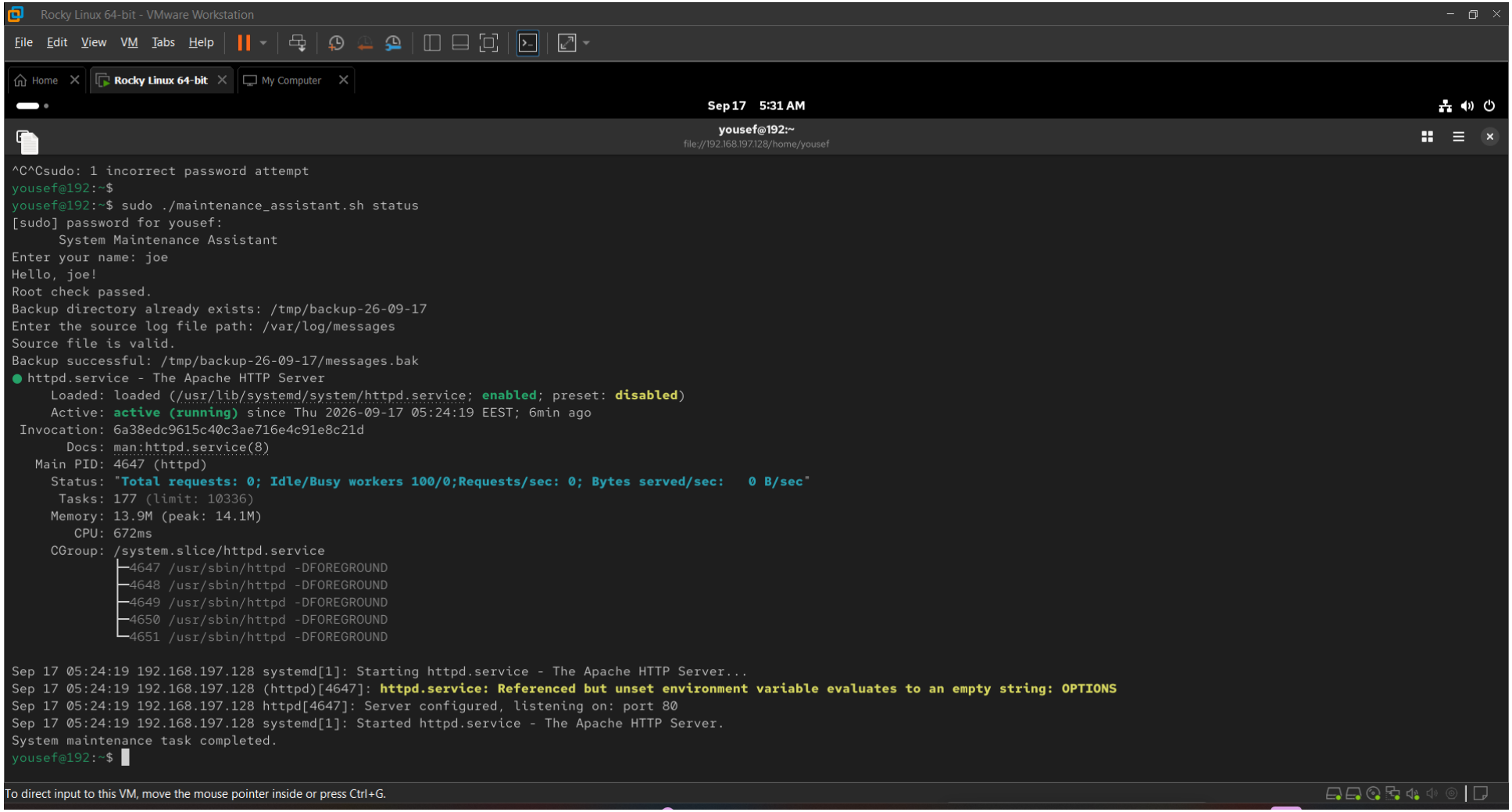
How the script works

The script reads and validates a name, greets the user, and checks that the effective user ID is 0 (root). It uses date to name a backup directory under /tmp and creates it only if it does not already exist.

It asks for a log file path, checks that the file exists and is not empty, then copies it with a .bak extension and checks the copy result. A case statement reads the first argument and calls systemctl to start, stop, restart, or show the status of httpd. An invalid or missing action displays the usage message and exits.

Successful execution

Command: sudo ./maintenance_assistant.sh status



The run validates /var/log/messages, creates messages.bak in the dated backup directory, shows httpd as active (running), and reaches the completion message. The entered name is joe.

Backup and validation evidence

Terminal screenshots from the Rocky Linux environment

Backup creation

The directory listing confirms messages.bak exists and has a size of 395K.

```
yousef@192:~$ ls -lh /tmp/backup-$(date +%y-%m-%d)
total 396K
-rw-------. 1 root root 395K Sep 17 05:30 messages.bak
yousef@192:~$
```

Root-check failure

Running without sudo displays the root requirement and returns to the prompt.

```
yousef@192:~$ ls -lh /tmp/backup-$(date +%y-%m-%d)
total 396K
-rw-------. 1 root root 395K Sep 17 05:30 messages.bak
yousef@192:~$ ./maintenance_assistant.sh status
      System Maintenance Assistant
Enter your name: joe
Hello, joe!
Error: This script must be run as root.
yousef@192:~$
```

Invalid-argument handling

The action hello reaches the default case and displays the allowed actions.

```
yousef@192:~$ sudo ./maintenance_assistant.sh hello
      System Maintenance Assistant
Enter your name: joe
Hello, joe!
Root check passed.
Backup directory already exists: /tmp/backup-26-09-17
Enter the source log file path: /var/log/messages
Source file is valid.
Backup successful: /tmp/backup-26-09-17/messages.bak
Usage: ./maintenance_assistant.sh {start|stop|restart|status}
yousef@192:~$
```